








Modelling Financial Markets

Seminar 20: Path Signatures and Realised Volatility — Hands-on



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Does the Shape of the Past Help Forecast Volatility?

- Questions of the seminar
  - **does the shape of the recent volatility path help to forecast realised variance?**
  - **how do we test that honestly?**
- Outputs to produce
  - answers to A1–A3 and the required independent task A6
  - one figure and one table for each of B1, B2, B3 and B4; optional B5 transfers the B1 analysis to another asset
  - the corrected C1 code
  - one interpretation sentence per exercise
- Notebook for today: open the seminar notebook in Colab
  - each task names its notebook section
- Exercises, types and order: the Exercise Map after the definitions
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What Do the Realised Measures Measure?

- VOLARE means VOLatility Archive for Realized Estimates. The database provides daily measures constructed from intraday prices; each asset keeps its own trading calendar.
- Realised Variance (RV) adds squared intraday returns. Realised Quarticity (RQ) uses fourth powers and helps assess how noisy the RV measurement is. The return channel uses open-to-close log returns, excluding overnight moves.
- JPM is the stock ticker of JPMorgan Chase; C2 uses all 40 stocks. ES denotes E-mini S&P 500 futures; CL denotes crude-oil futures. Here ES is a contract code, not Expected Shortfall.
- Foreign Exchange (FX) denotes currency markets. Currency pairs use EUR (euro), GBP (British pound), AUD (Australian dollar), USD (US dollar), CAD (Canadian dollar) and JPY (Japanese yen), in base/quote order.

Data Used

Data set	Assets	Sample	Used in
VOLARE realised measures, US stocks	JPM (all 40 stocks in C2)	2 Jan 2015 – 30 Jan 2026	A7, B1–B3, C2
VOLARE realised measures, futures	ES (E-mini S&P 500), CL (crude oil)	28 Sep 2009 – 30 Jan 2026	B5–B7
Chapter 20 results (provided with the notebook)	50 assets	one-sided Diebold–Mariano (DM) p-values	B4, B8

- Variables of the VOLARE data Cipollini et al. (2026), one value per asset and trading day
  - Five-minute realised variance RV, its semivariances RV_t^+, RV_t^- and realised quarticity RQ (definitions: Realised Variance, Semivariances and Quarticity)
  - The daily open and close prices, for the open-to-close log return r_t
- VOLARE is free for research at volare.unime.it: download the realised-variance data (stocks, futures, FX) and point the first notebook cell to them

Setup: Run and Verify

```
# notebook section "Signatures, models and tests"
# defines load_asset(kind, symbol): columns
# rv, rsp, rsn, rq and r = ln(close/open)
a = load_asset('stocks', 'JPM')
print(len(a), a.index[0].date(), a.index[-1].date())
print(pd.DataFrame({'RV x 1e4': 1e4 * a.rv,
                    'r (%)': 100 * a.r}).head(3))
```

- Expected output: 2 786 trading days, 2015-01-02 to 2026-01-30

date	RV $\times 10^4$	r (%)
2015-01-02	1.463	-0.22
2015-01-05	1.855	-2.56
2015-01-06	2.974	-2.77

- RV of 1.46×10^{-4} is a daily volatility of $\sqrt{1.46 \times 10^{-4}} = 1.2\%$
- If your count or first value differs, check the data set and the asset

Why Does the Time Scale Matter?

- The hand calculations A1–A6 use time in days: each daily step contributes one unit to the time channel. This keeps the geometry of the examples easy to check.
- The empirical features of Chapter 20 instead rescale the time channel of every window to the interval from zero to one. A 22-observation window then has 21 equal time steps.
- Changing this scale changes every signature term containing a time coordinate. A numerical area from the hand exercise cannot be compared directly with an empirical feature built on the rescaled path.
- Keep the same convention when comparing paths or computing kernel distances. Rescale before computing the signature, not after mixing features from different conventions.

A Window of Data Is a Path

- Details (theorems, the full design, the 50-asset results): Chapter 20
- Take the last L days of a few series and join the points by straight lines: a path

$$X : [a, b] \to \mathbb{R}^d, \qquad X = \left(t,\ \ln \mathrm{RV}_t,\ \textstyle\sum_{s \le t} r_s\right)$$

  - L: window length in days (22 in the exercises); d: number of channels (coordinates), here $d = 3$; $[a, b]$: the time interval of the window
  - RV_t: realised variance of day t; r_s: open-to-close log return of day s, so the third channel cumulates intraday log returns, excluding overnight moves
- **Time augmentation**: channel 1 is time itself
  - Without it a fast move and a slow move with the same values look the same
- Increments: $\Delta_k = X_{t_k} - X_{t_{k-1}} \in \mathbb{R}^d$, the move of all channels over segment k
  - Δ_k^i: component i of Δ_k; for the time channel $\Delta_k^{\mathbf{1}} = 1$ day

The Signature

- **Signature**: all iterated integrals of the path, collected by level k Chevyrev & Kormilitzin (2016)

$$S^{i_{\mathbf{1}}\cdots i_k} = \int_{a<u_{\mathbf{1}}<\cdots<u_k<b} dX^{i_{\mathbf{1}}}_{u_{\mathbf{1}}} \cdots dX^{i_k}_{u_k}$$

  - $(i_{\mathbf{1}}, \ldots, i_k)$: a word, an ordered list of k channels from $\{1, \ldots, d\}$; $u_{\mathbf{1}} < \cdots < u_k$: ordered times; dX^i_u: move of channel i at time u

- Level 1: the total increments; level 2: the $d \times d$ matrix of moves of j weighted by how far i has already moved

$$S^i = X^i_b - X^i_a, \qquad S^{ij} = \int_a^b (X^i_u - X^i_a)\, dX^j_u$$

  - $S^{(\mathbf{1})}$, $S^{(\mathbf{2})}$: level 1 (a vector) and level 2 (a matrix); one straight segment with increment Δ: $S^{(\mathbf{1})} = \Delta$, $S^{(\mathbf{2})} = \frac{\mathbf{1}}{\mathbf{2}}\Delta\Delta^\top$
  - Depth N: levels $1, \ldots, N$ are kept (truncated signature Sig^N); level k has d^k terms

- Why use it: with a strictly increasing time channel the full signature determines the path up to its starting point Hambly & Lyons (2010), and linear functions of signatures of sufficiently high depth approximate continuous path functions on compact sets with a common starting point, so it serves as a set of regression features

Chen's Identity and the Shuffle Identity

- **Chen's identity** Chen (1957): the signature of two joined paths is the tensor product of their signatures; at level 2

$$S^{(2)}(X * Y) = S^{(2)}(X) + S^{(1)}(X)\, S^{(1)}(Y)^\top + S^{(2)}(Y)$$

  - $X * Y$: path X followed by path Y (joined end to start); $S^{(1)}(X)\, S^{(1)}(Y)^\top$: the $d \times d$ matrix of products, every move of X comes before every move of Y

- Applied segment by segment to a path with increments $\Delta_1, \ldots, \Delta_m$

$$S^{(2)} = \sum_{k=1}^{m} \tfrac{1}{2} \Delta_k \Delta_k^\top + \sum_{j<k} \Delta_j \Delta_k^\top$$

  - First sum: each segment with itself; second sum: every ordered pair of segments, j before k

- **Shuffle identity**: a product of level-1 terms is a sum of level-2 terms

$$S^i S^j = S^{ij} + S^{ji}$$

  - So the symmetric part of level 2 is $\tfrac{1}{2}\Delta X^i \Delta X^j$, with $\Delta X^i = S^i$ the total change of channel i: it brings nothing new

The Lévy Area

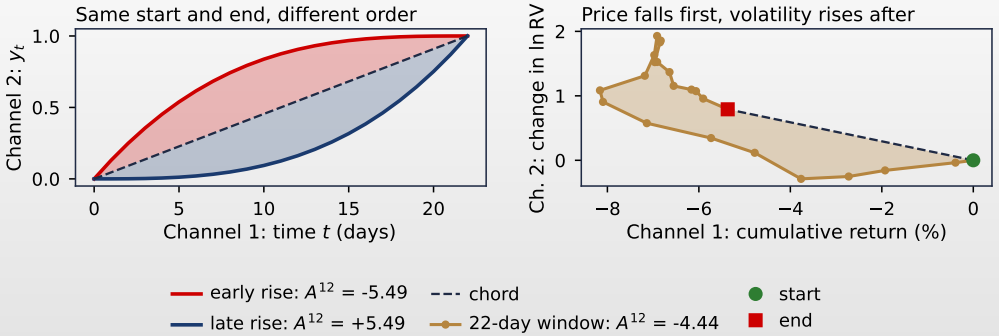
- **Lévy area**: the antisymmetric part of level 2

$$A^{ij} = \tfrac{1}{2}\left(S^{ij} - S^{ji}\right)$$

  - The signed area between the path (projected on channels i, j) and its chord, the straight line from start to end; anticlockwise > 0, clockwise < 0
  - It records the **order** of the moves: $A^{ij} > 0$ when channel i tends to move before channel j

- Small example, channels (time, y), $y = (0, 1, 1)$: $\Delta_{\mathbf{1}} = (1, 1)$, $\Delta_{\mathbf{2}} = (1, 0)$
  - $S^{(\mathbf{1})} = (2, 1)$; $S^{(\mathbf{2})} = \tfrac{1}{2}\Delta_{\mathbf{1}}\Delta_{\mathbf{1}}^{\top} + \tfrac{1}{2}\Delta_{\mathbf{2}}\Delta_{\mathbf{2}}^{\top} + \Delta_{\mathbf{1}}\Delta_{\mathbf{2}}^{\top} = \left[\begin{smallmatrix} \mathbf{2} & \mathbf{0.5} \\ \mathbf{1.5} & \mathbf{0.5} \end{smallmatrix}\right]$; shuffle: $S^{\mathbf{1}}S^{\mathbf{2}} = 2 = 0.5 + 1.5$
  - $A^{\mathbf{12}} = (0.5 - 1.5)/2 = -0.5$: y rose early; for $y = (0, 0, 1)$ (late rise) the same steps give $A^{\mathbf{12}} = +0.5$

The Lévy Area on Simulated Paths



- Left: two 22-day paths with the same start and end; the shaded region lies between path and chord
  - early rise: $A^{12} = -5.49$ (clockwise); late rise: $A^{12} = +5.49$; level 1 is identical, only the area tells them apart
- Right: channels (cumulative return, $\ln \mathrm{RV}$); the price falls first, volatility rises afterwards: $A^{12} = -4.44$
  - the kind of order effect that B1 tests as a predictor of future volatility

What Does Each Forecasting Model Add?

- HAR means Heterogeneous AutoRegressive: recent variance is summarised over daily, weekly and monthly horizons. Log-HAR models log variance; it is the main benchmark here.
- SHAR is semivariance HAR: positive and negative intraday moves enter separately. HARQ adds realised quarticity to account for variation in the precision of RV.
- LASSO means Least Absolute Shrinkage and Selection Operator. Its penalty can set coefficients to zero and select features; Ordinary Least Squares (OLS) then refits the selected set without that penalty.
- Sig-LK is signature LASSO with kernel weights. It adds path features and gives more weight to similar past windows; the question is whether these additions improve forecasts beyond log-HAR.

Sig-LK, Features and Kernel Weights

- Signature LASSO with kernel weights of Gu et al. (2024), applied to next-day (or next-week, next-month) log RV; benchmark: log-HAR (next slides)
- Features ξ_τ of day τ: the three log-HAR terms and the signature up to depth 3 of the 22-day path (time, $\ln \mathrm{RV}$, cumulative r) ending at day τ
  - τ: a past day of the training window; t: the forecast origin, the last day observed
- Kernel weights: past days whose window looks like the latest one get more weight

$$w_\tau = \frac{e^{-\gamma \delta_\tau}}{\sum_s e^{-\gamma \delta_s}}, \qquad \gamma = \frac{c}{\mathrm{median}(\delta)}$$

  - δ_τ: squared Euclidean distance between the truncated signatures of the window ending at τ and of the window ending at t; s: runs over all training days, so the weights sum to 1
  - $\gamma \geq 0$: inverse temperature (called temperature in the paper); $c > 0$: a unit-free constant; with $c = 1$ a window at the median distance gets weight $e^{-1} = 0.37$ relative to an identical one; $\gamma = 0$ gives equal weights

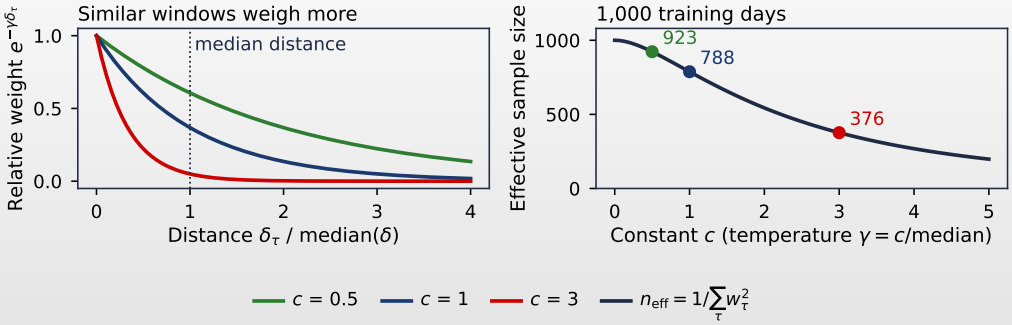
Sig-LK, Effective Sample Size and the Two-Step LASSO

- Cost of unequal weights: fewer effective observations (Kish; A3)

$$n_{\text{eff}} = \frac{1}{\sum_\tau w_\tau^2}$$

  - $n_{\text{eff}} = n$ for equal weights $1/n$; $n_{\text{eff}} = 1$ if one day takes all the weight
- Two estimation steps on the weighted training sample
  - Step 1: a weighted LASSO minimises $\sum_\tau w_\tau (y_\tau - \xi_\tau^\top \theta)^2 + \lambda \sum_j |\theta_j|$; y_τ: the log-RV target, θ: coefficients; the penalty $\lambda \geq 0$ sets some coefficients exactly to zero, i.e. selects the features
  - λ is chosen by BIC (Bayesian information criterion: fit plus a penalty per kept coefficient)
  - Step 2: weighted OLS refits the selected features only, removing the LASSO shrinkage towards zero
- Rolling design: the last 1000 complete pairs (features, target), re-estimated every 5 days; each forecast uses only data up to its origin (no look-ahead)

Kernel Weights and the Effective Sample Size



- Left: relative weight $e^{-\gamma\delta_\tau}$ of a past window at distance δ_τ, with $\gamma = c/\text{median}(\delta)$
  - at the median distance: $e^{-\mathbf{0.5}} = 0.61$, $e^{-\mathbf{1}} = 0.37$, $e^{-\mathbf{3}} = 0.05$ for $c = 0.5, 1, 3$
- Right: 1 000 simulated distances; $n_{\text{eff}} = 923, 788, 376$ for $c = 0.5, 1, 3$
  - larger inverse temperature: more similar days, but fewer effective observations (A3)

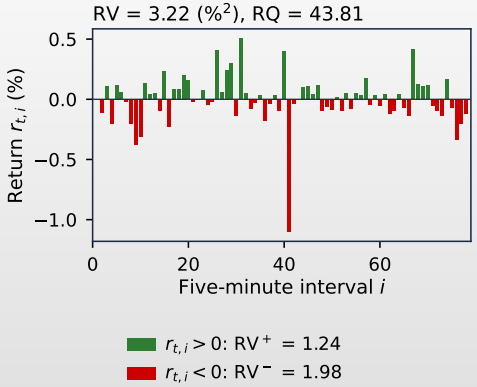
Which Claim Does Each Test Support?

- Quasi-likelihood (QLIKE) scores variance forecasts. Diebold–Mariano (DM) tests whether two models have equal average loss; conditional Giacomini–White (GW) also uses information available before the target.
- HAC means Heteroskedasticity and Autocorrelation Consistent. It estimates uncertainty while allowing unequal variances and temporal dependence, including the dependence caused by overlapping targets.
- Holm controls the chance of any false rejection in the tested family. Benjamini–Hochberg (BH) instead controls the expected share of false discoveries under its dependence assumptions.
- A small p-value is evidence about the specified null, not a measure of the economic gain. Compare the loss ratio, its confidence interval and the relevant test together.

Tools from Earlier Chapters, Task by Task

- B1, B2: HAR and log-HAR Corsi (2009), SHAR with semivariances Patton & Sheppard (2015), HARQ with realised quarticity Bollerslev, Patton & Quaedvlieg (2016), Newey–West (HAC) standard errors: Chapter 9
- A2, B3: QLIKE and why it ranks variance forecasts robustly to a noisy proxy Patton (2011): Chapter 9
- B3: Diebold–Mariano Diebold & Mariano (1995) and conditional Giacomini–White Giacomini & White (2006) tests: Chapter 8; stationary bootstrap for an interval: Chapter 5
- B4: Holm (FWER) and Benjamini–Hochberg (FDR) corrections Benjamini & Hochberg (1995): Chapter 3
- New today are only the signature, the Lévy area and the kernel weights of the previous slides; the next slides recall the formulas of these earlier tools, with simulated examples

Realised Variance, Semivariances and Quarticity



- Day t split into M intraday returns $r_{t,i}$ ($M = 78$ five-minute intervals)

$$\mathrm{RV}_t = \sum_{i=1}^{M} r_{t,i}^2, \qquad \mathrm{RV}_t^+ = \sum_{i=1}^{M} r_{t,i}^2 \, 1\{r_{t,i} > 0\}$$

  - RV_t estimates the variance of day t; $\sqrt{\mathrm{RV}_t}$: the daily volatility
  - $\mathrm{RV}_t^- = \mathrm{RV}_t - \mathrm{RV}_t^+$: "bad" volatility from falling prices (SHAR)
- Realised quarticity: the scale of the measurement error of RV_t (HARQ)

$$\mathrm{RQ}_t = \frac{M}{3} \sum_{i=1}^{M} r_{t,i}^4$$

  - large on days with a few big moves; the error variance of RV_t is about $2\,\mathrm{RQ}_t / M$
- Chart: one simulated day with one jump; $\mathrm{RV} = 3.22$, $\mathrm{RV}^- = 1.98 > \mathrm{RV}^+ = 1.24$

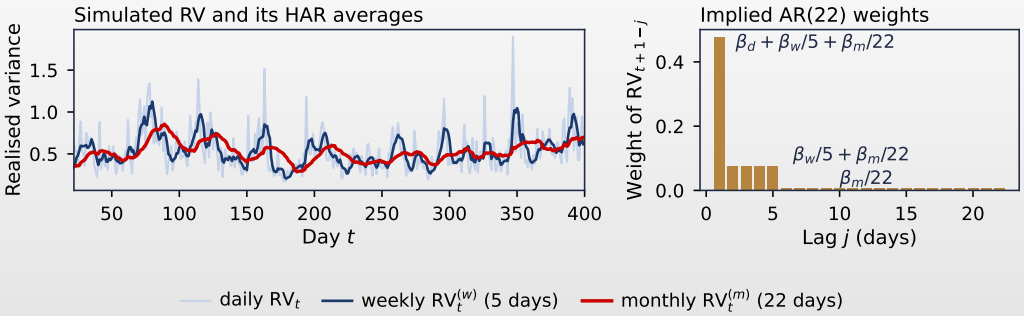
Recall, the HAR Family (B1, B2)

- HAR Corsi (2009): next-period RV is a linear function of the daily, weekly and monthly averages of past RV

$$\mathrm{RV}_{t+\mathbf{1}} = \beta_{\mathbf{0}} + \beta_d \,\mathrm{RV}_t + \beta_w \,\mathrm{RV}_t^{(w)} + \beta_m \,\mathrm{RV}_t^{(m)} + \varepsilon_{t+\mathbf{1}}$$

  - $\mathrm{RV}_t^{(w)}$, $\mathrm{RV}_t^{(m)}$: averages of RV over the last 5 and 22 days; $\beta_d, \beta_w, \beta_m$: weights of yesterday, last week, last month; $\varepsilon_{t+\mathbf{1}}$: error; log-HAR: the same in logs
- SHAR Patton & Sheppard (2015): the daily term split into the semivariances RV_t^+ (from positive intraday returns) and RV_t^- (from negative ones)
  - Coefficients β^+, β^- instead of β_d; $\beta^- > \beta^+$: bad volatility predicts more future volatility than good volatility
- HARQ Bollerslev, Patton & Quaedvlieg (2016): the weight of RV_t falls on days when RV is measured less precisely
  - Adds $\beta_Q \,\mathrm{RV}_t(\mathrm{RQ}_t^{\mathbf{1/2}} - \overline{\mathrm{RQ}^{\mathbf{1/2}}})$ to HAR; RQ_t: realised quarticity (Realised Variance, Semivariances and Quarticity); bar: sample mean; expected $\beta_Q < 0$

The HAR Averages and Their Lag Weights



- Left: a simulated RV series with its weekly and monthly averages; the monthly one moves slowly
- Right: HAR is an autoregressive model of order 22, AR(22) with weights restricted to three steps (A7); here $\beta_d = 0.4$, $\beta_w = 0.35$, $\beta_m = 0.15$
  - lag 1: 0.477, lags 2–5: 0.077 each, lags 6–22: 0.0068 each; 4 parameters instead of 23

Newey–West (HAC) Standard Errors (B1, B2, B3)

- Regression $y_t = x_t^\top \beta + u_t$ with heteroskedastic, autocorrelated errors; long-run covariance of $x_t u_t$

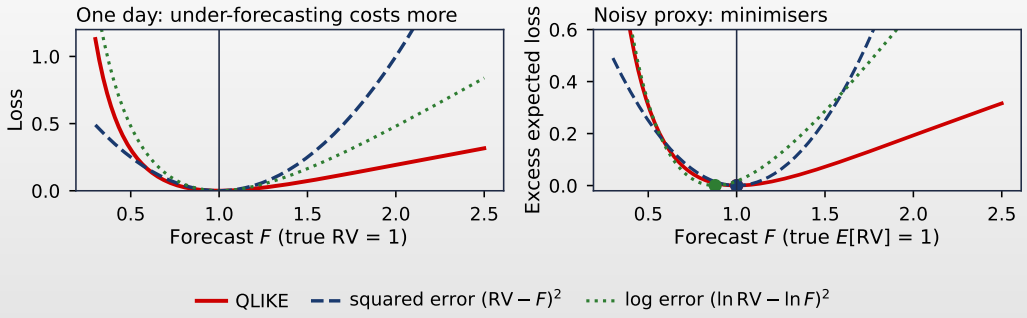
$$\hat{\Omega} = \hat{\Gamma}_0 + \sum_{l=1}^{q} \left(1 - \frac{l}{q+1}\right)(\hat{\Gamma}_l + \hat{\Gamma}_l^\top), \qquad \hat{\Gamma}_l = \frac{1}{T} \sum_{t=l+1}^{T} \hat{u}_t \hat{u}_{t-l}\, x_t x_{t-l}^\top$$

  - q: number of lags (bandwidth); $1 - l/(q+1)$: Bartlett weights, which keep $\hat{\Omega}$ positive semidefinite
- The sandwich covariance of the OLS estimates and the t statistics

$$\widehat{\mathrm{Var}}(\hat{\beta}) = (X^\top X)^{-1}\, T\hat{\Omega}\, (X^\top X)^{-1}, \qquad t = \frac{\hat{\beta}_j}{\mathrm{se}(\hat{\beta}_j)} \approx N(0,1)$$

  - for a difference of two coefficients (B2): $\mathrm{se}(\hat{\beta}^- - \hat{\beta}^+) = \sqrt{V_{--} + V_{++} - 2V_{+-}}$
  - overlapping h-day targets give u_t a Moving Average (MA) structure of order $(h-1)$: take $q \geq h-1$ (22 lags for monthly targets in B1)
- $R^2 = 1 - \sum_t \hat{u}_t^2 / \sum_t (y_t - \bar{y})^2$: share of the variance of y explained by the regressors

QLIKE and a Noisy Volatility Proxy (A2, B3)



- $\mathrm{QLIKE}(\mathrm{RV}, F) = \mathrm{RV}/F - \ln(\mathrm{RV}/F) - 1$: zero only at $F = \mathrm{RV}$; left: it rises steeply when F is too low
- Right: the proxy is noisy ($\mathrm{RV} = \chi^2_8/8$, mean 1); QLIKE and squared error are minimised at $F = \mathbb{E}[\mathrm{RV}] = 1.00$, the log error at 0.88
  - a robust loss Patton (2011) ranks forecasts as the true variance would; the log error rewards a biased forecast

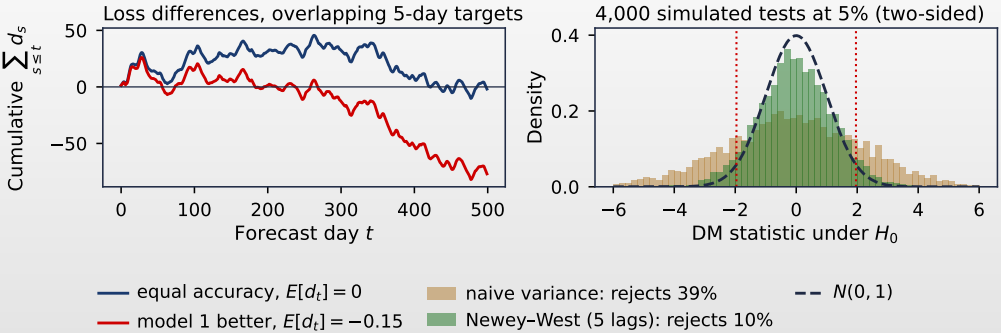
Recall, Comparing Two Forecasts (B3)

- Diebold–Mariano test Diebold & Mariano (1995): is the mean loss difference zero?

$$d_t = L_t^{(\mathbf{1})} - L_t^{(\mathbf{2})}, \qquad \mathrm{DM} = \frac{\bar{d}}{\sqrt{\hat{\Omega}_{\mathrm{NW}}/T}} \approx N(0,1) \text{ under } H_{\mathbf{0}}\colon \mathbb{E}[d_t] = 0$$

  - $L_t^{(\mathbf{1})}$, $L_t^{(\mathbf{2})}$: losses (e.g. QLIKE) of models 1 and 2 on day t; $\bar{d}$: mean of d_t over the T forecast days; $\hat{\Omega}_{\mathrm{NW}}$: Newey–West (NW) long-run variance of d_t
  - $\mathrm{DM} < -1.645$: model 1 significantly more accurate at 5% (one-sided)
- Conditional Giacomini–White test Giacomini & White (2006): can the loss difference be predicted from past information?
  - For a one-day target, regress d_t on instruments $(1, d_{t-\mathbf{1}})$; Wald statistic for both coefficients $= 0$, approximately $\chi^{\mathbf{2}}(2)$ under $H_{\mathbf{0}}$ (2 = number of instruments)
- Stationary bootstrap: resample blocks of consecutive days of random length (mean block length set by the user) to keep the serial dependence
  - The 2.5% and 97.5% quantiles of the replicated statistic give a 95% CI (confidence interval)

The Diebold–Mariano Test with Overlapping Targets



- Simulated loss differences d_t of 5-day forecasts: neighbouring days share 4 of 5 days, so d_t is autocorrelated
  - left: the cumulative sum drifts down only when model 1 is truly better
  - right: under H_0 the naive variance rejects 39% of the time instead of 5%; Newey–West brings it to 10%

Recall, Many Tests at Once (B4)

- m tests (here one per asset) with p-values sorted increasingly, $p_{(1)} \leq \cdots \leq p_{(m)}$
  - At 5% each, about 0.05 m rejections are expected by chance even if no model is better (2.5 of 50)
- **Holm** controls the FWER (family-wise error rate): the probability of at least one false rejection is at most 5%

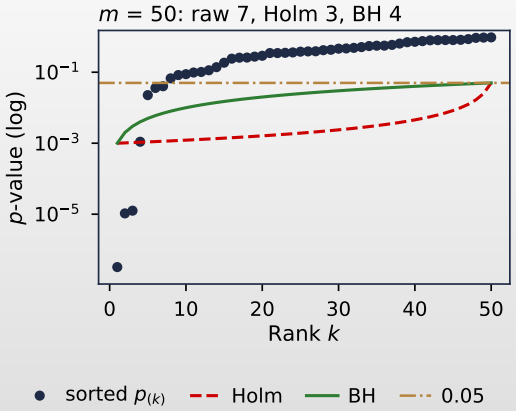
$$\text{reject while } p_{(k)} \leq \frac{0.05}{m-k+1},\ k = 1, 2, \ldots;\ \text{stop at the first failure}$$

- **Benjamini–Hochberg (BH)** Benjamini & Hochberg (1995) controls the FDR (false discovery rate): the expected share of false rejections among all rejections is at most 5%

$$k^* = \max\left\{k : p_{(k)} \leq 0.05\,\frac{k}{m}\right\};\quad \text{reject the } k^* \text{ smallest } p\text{-values}$$

  - Holm is valid under any dependence; BH needs independent or positively dependent p-values; BH rejects at least as many as Holm

Holm and BH on 50 Simulated Tests



- 50 simulated one-sided tests, 5 with a true effect
  - dots: sorted $p_{(k)}$; a point below a line is rejected by that rule
- Rejections: unadjusted 7, Holm 3, BH 4
  - with $m = 50$ the first Holm threshold is $0.05/50 = 0.001$: only very small p-values survive
  - of the 7 unadjusted rejections, some are false: about $0.05 \cdot 45 = 2.25$ are expected from the 45 true nulls

Notation Used in Every Exercise

- RV_t: five-minute realised variance of day t; $r_t = \ln(\text{close}_t/\text{open}_t)$: open-to-close log return
- HAR averages: $\mathrm{RV}_t^{(w)} = \frac{1}{5}\sum_{j=0}^{4}\mathrm{RV}_{t-j}$, $\mathrm{RV}_t^{(m)} = \frac{1}{22}\sum_{j=0}^{21}\mathrm{RV}_{t-j}$; log-HAR regresses $\ln \mathrm{RV}_{t+1:t+h}$ on $\ln \mathrm{RV}_t$, $\ln \mathrm{RV}_t^{(w)}$, $\ln \mathrm{RV}_t^{(m)}$, with $\mathrm{RV}_{t+1:t+h}$ the mean RV over the next h days
- Defined in Concepts and Notation
  - Path, increments Δ_k, signature levels $S^{(1)}$, $S^{(2)}$ and Lévy area A^{12}: see slides "A Window of Data Is a Path" to "The Lévy Area"
  - Kernel weights w_τ: Sig-LK, Features and Kernel Weights; effective sample size n_{eff}: Sig-LK, Effective Sample Size and the Two-Step LASSO
  - QLIKE loss $\mathrm{QLIKE}(\mathrm{RV}, F)$: QLIKE and a Noisy Volatility Proxy (A2, B3); DM statistic of the loss difference d_t: Recall, Comparing Two Forecasts (B3)
- Worked example: $\mathrm{RV} = 1$, $F = 2$: $\mathrm{QLIKE} = 0.5 - \ln 0.5 - 1 = 0.193$

Exercise Map

Exercise	Question	Type	Model
A1	how does a signature record the order of moves?	Solved	–
A2	which forecast minimises QLIKE, which errors cost more?	Solved	–
A3	how does the temperature trade similarity for sample size?	Solved	–
B1	does the Lévy area predict next month's RV of JPM?	Solved	–
B2	do the SHAR and HARQ effects hold for JPM?	Solved	–
B3	does Sig-LK beat log-HAR for JPM, one day ahead?	Solved	–
B4	how many wins survive Holm and BH over 50 assets?	Solved	–
C1	what is wrong in an artificial intelligence (AI) answer?	Proposed	A1, A2, B3
Extension packs	A4–A5, A7–A10, B5–B8, C2 (optional); A6 required	Solved and Proposed (marked on each task)	named on each task

- **[Solved]**: full solution in the slides and in the notebook, a model to copy; **[Proposed]**: you solve it, the solution is discussed in class
- Part A: derivations on paper; Part B: verification and inference on VOLARE data, each ending with an interpretation question; Part C: open questions and AI critique

A1 [Solved]: A Depth-2 Signature, Step by Step — Task

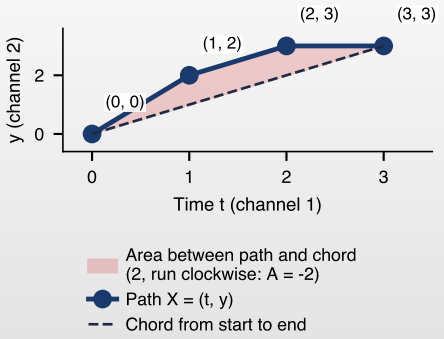
- **Question**: how does a depth-2 signature record *when* a series moved, not only how much?
- **Inputs**
  - A series takes the values $y = (0, 2, 3, 3)$ on days $t = 0, 1, 2, 3$; no data file
  - The time-augmented path $X_t = (t, y_t)$ is linear between consecutive days (channel 1 = time, channel 2 = y)
- **Tasks**
  (a) Write the three increments $\Delta_1, \Delta_2, \Delta_3 \in \mathbb{R}^2$ and compute level 1, $S^{(1)} = \sum_k \Delta_k$.
  (b) Use Chen's identity to derive the level-2 formula (see Chen's Identity and the Shuffle Identity) (squares plus ordered pairs), then compute this 2×2 matrix.
  (c) Check the result with the shuffle identity (see Chen's Identity and the Shuffle Identity) for $i = j = 1$ and for $i = 1, j = 2$.
  (d) Compute the Lévy area A^{12} (The Lévy Area) and draw the path with its chord.
- **Report**: $S^{(1)}$, $S^{(2)}$ and A^{12}
- **Interpretation**: what does the sign of A^{12} say about when the series rose?
- Notebook: section A1

A1 [Solved]: Solution (1/2) — Levels 1 and 2

- (a) Increments: $\Delta_1 = (1, 2)$, $\Delta_2 = (1, 1)$, $\Delta_3 = (1, 0)$; level 1: $S^{(1)} = (3, 3)$, the total change in time and in y
- (b) One segment has $\mathrm{Sig} = (1, \Delta, \frac{1}{2}\Delta\Delta^\top, \dots)$; Chen multiplies the segments, and level 2 of the product collects every square term once and every ordered pair $j < k$ once
- Squares: $\frac{1}{2}\begin{bmatrix}1 & 2\\ 2 & 4\end{bmatrix} + \frac{1}{2}\begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix} + \frac{1}{2}\begin{bmatrix}1 & 0\\ 0 & 0\end{bmatrix} = \begin{bmatrix}1.5 & 1.5\\ 1.5 & 2.5\end{bmatrix}$
- Ordered pairs: $\Delta_1\Delta_2^\top + \Delta_1\Delta_3^\top + \Delta_2\Delta_3^\top = \begin{bmatrix}1 & 1\\ 2 & 2\end{bmatrix} + \begin{bmatrix}1 & 0\\ 2 & 0\end{bmatrix} + \begin{bmatrix}1 & 0\\ 1 & 0\end{bmatrix} = \begin{bmatrix}3 & 1\\ 5 & 2\end{bmatrix}$
- Sum: $S^{(2)} = \begin{bmatrix}4.5 & 2.5\\ 6.5 & 4.5\end{bmatrix}$

A1 [Solved]: Solution (2/2) — Shuffle Check and Lévy Area

- (c) Shuffle, $i = j = 1$: $S^1S^1 = 9 = 2S^{11} = 2 \cdot 4.5$; $i = 1, j = 2$: $S^1S^2 = 9 = S^{12} + S^{21} = 2.5 + 6.5$: both hold
- (d) Lévy area: $A^{12} = (2.5 - 6.5)/2 = -2$
- Geometric check (chart): the path lies above its chord $y = t$; the shaded polygon has area 2 and is run clockwise, hence the minus sign
- Interpretation: with channels (time, y), a negative area means that y rose *early* in the window; the same net change spread evenly would give $A^{12} = 0$



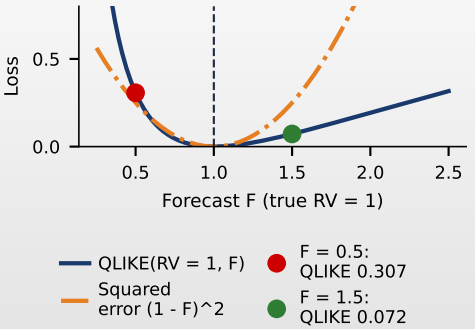
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A2 [Solved]: QLIKE: Minimiser and Asymmetry — Task

- **Questions**
  - which forecast minimises expected QLIKE?
  - which errors does QLIKE punish more?
- **Inputs**: $\mathrm{QLIKE}(\mathrm{RV}, F)$ of QLIKE and a Noisy Volatility Proxy (A2, B3), with realised proxy $\mathrm{RV} > 0$ and forecast $F > 0$; $\mathbb{E}_t$ = expectation given the information at the forecast origin
- **Tasks**
  (a) Differentiate $\mathbb{E}_t[\mathrm{QLIKE}(\mathrm{RV}, F)]$ with respect to F and solve the first-order condition.
  (b) Check the second-order condition.
  (c) For $\mathrm{RV} = 1$, compute QLIKE and the squared error for $F = 0.5$ and for $F = 1.5$, and plot both losses against F.
- **Report**: the minimiser, the four losses and the chart
- **Interpretation**
  - which error does QLIKE punish more?
  - why does this suit risk management?
- Notebook: section A2

A2 [Solved]: Solution

- (a) $\mathbb{E}_t[\mathrm{QLIKE}] = \mathbb{E}_t[\mathrm{RV}]/F + \ln F - \mathbb{E}_t[\ln \mathrm{RV}] - 1$; $\partial/\partial F = -\mathbb{E}_t[\mathrm{RV}]/F^{\mathbf{2}} + 1/F = 0 \Rightarrow F^* = \mathbb{E}_t[\mathrm{RV}]$
- (b) $\partial^{\mathbf{2}}/\partial F^{\mathbf{2}} = 2\mathbb{E}_t[\mathrm{RV}]/F^{\mathbf{3}} - 1/F^{\mathbf{2}} = 1/F^{*\mathbf{2}} > 0$ at F^*: a minimum; F^* depends only on $\mathbb{E}_t[\mathrm{RV}]$, so with a conditionally unbiased proxy the *expected*-loss ranking is robust Patton (2011)
- (c) $\mathrm{QLIKE}(1, 0.5) = 2 - \ln 2 - 1 = 0.307$; $\mathrm{QLIKE}(1, 1.5) = 0.667 + \ln 1.5 - 1 = 0.072$; squared errors 0.25 and 0.25
- Interpretation: QLIKE punishes under-prediction of variance about four times more than the same over-prediction; under-predicted risk is the costly error (too little capital, breached VaR)

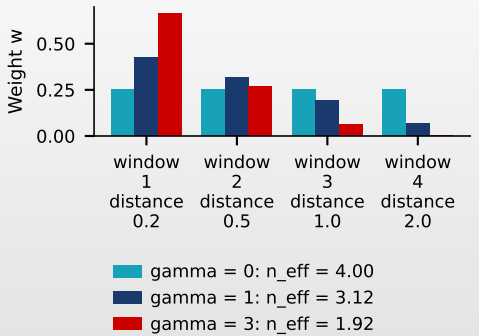


A3 [Solved]: Kernel Weights and Effective Sample Size — Task

- **Question**: how does the kernel inverse temperature γ trade similarity against the number of observations?
- **Inputs**
  - Four past windows with signature distances $\delta = (0.2, 0.5, 1.0, 2.0)$ to the current window
  - Weights w_τ of Gu et al. (Eq. 12; Sig-LK, Features and Kernel Weights) and the Kish effective sample size n_{eff} (Sig-LK, Effective Sample Size and the Two-Step LASSO)
- **Tasks**
  (a) Compute the four weights for $\gamma = 1$.
  (b) Compute the four weights for $\gamma = 3$.
  (c) Compute n_{eff} in both cases and compare with equal weights ($\gamma = 0$).
- **Report**: two weight vectors, two n_{eff}, a bar chart
- **Interpretation**
  - what is gained when γ increases?
  - what is lost?
- Notebook: section A3

A3 [Solved]: Solution

- (a) $e^{-\delta} = (0.819, 0.607, 0.368, 0.135)$, sum 1.929: $w = (0.425, 0.315, 0.191, 0.070)$
- (b) $e^{-\mathbf{3}\delta} = (0.549, 0.223, 0.050, 0.002)$, sum 0.824: $w = (0.666, 0.271, 0.060, 0.003)$
- (c) $n_{\text{eff}} = 4$ for $\gamma = 0$, 3.12 for $\gamma = 1$ and 1.92 for $\gamma = 3$
- Interpretation: a larger inverse temperature focuses on the most similar window (less bias from dissimilar periods) but halves the information (more variance); for log-HAR-K in the lecture $n_{\text{eff}} \approx 816$ of 1000: little is lost, and little changes



B1 [Solved]: Does the Lévy Area Predict Volatility? (JPM) — Task

- **Question**: does the order of price and volatility moves over the last month predict JPM's volatility next month, beyond log-HAR?
- **Inputs**
  - VOLARE Cipollini et al. (2026), JPM (JPMorgan Chase), 2 Jan 2015 – 30 Jan 2026, trading days only; five-minute RV, daily open and close prices
  - Regressor: A_t = Lévy area of the 22-day path (cumulative r, $\ln \mathrm{RV}$) ending at day t, divided by its full-sample standard deviation; channel order as written, no time channel
  - Target: $y_t = \ln \mathrm{RV}_{t+\mathbf{1}:t+\mathbf{22}} - \ln \mathrm{RV}_t^{(m)}$, next month's log RV minus the past month's (consecutive targets overlap in 21 of 22 days)
- **Tasks**
  (a) For every day, build the path of the last 22 days, compute its depth-2 signature and the Lévy area A_t (The Lévy Area), then standardise A_t.
  (b) Regress y_t by OLS on a constant, the three log-HAR terms and A_t, with Newey–West standard errors and 22 lags.
  (c) Test H_0: coefficient of $A_t = 0$ at 5%, two-sided, and compare the R^2 with and without A_t.
  (d) Draw a binned scatter of y_t against A_t, both residualised on the log-HAR terms.
- **Report**: coefficient, HAC s.e., t, decision, both R^2, the chart
- **Interpretation**
  - why 22 lags?
  - is the gain economically relevant?
- Notebook: section B1

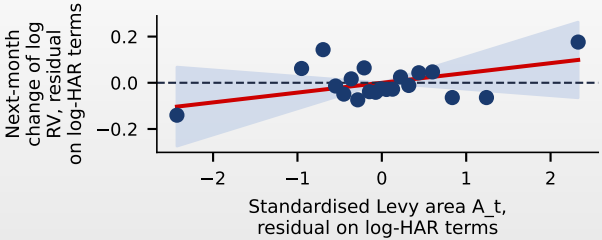
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B1 [Solved]: Solution (1/2) — Steps

- (a) Signature of each 22-point window with channels (return, $\ln \mathrm{RV}$); A_t is negative in 46% of the windows (compatible with, but not identifying, leverage-type sequences)
- (b) 2 743 usable days; coefficient of $A_t = +0.042$, HAC s.e. 0.036
- (c) $t = +0.042/0.036 = +1.18$ (from unrounded estimates); $|t| < 1.96$: do not reject H_0 at 5%; R^2: 28.0% without A_t, 28.6% with A_t
- Why 22 lags: consecutive targets share 21 of their 22 days, so the errors are serially dependent even under H_0; persistent RV can extend the dependence, so a check with 44 lags is advisable

B1 [Solved]: Solution (2/2) — Chart and Interpretation

- (d) Chart: bins scatter around zero; the HAC band of the slope contains the horizontal line
- Relevance: half a point of R^2 next to the 28% explained by the log-HAR terms is negligible
- Interpretation: for JPM the order of moves between price and volatility adds no reliable information about next month's volatility



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B2 [Solved]: HAR, SHAR and HARQ for JPM — Task

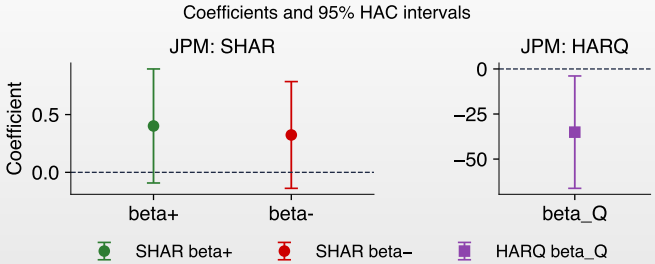
- **Question**: do bad volatility (SHAR) and the measurement error of RV (HARQ) change next-day forecasts for JPM?
- **Inputs**
  - VOLARE Cipollini et al. (2026), JPM (JPMorgan Chase), 2 Jan 2015 – 30 Jan 2026, trading days only; five-minute RV, its semivariances RV_t^+, RV_t^- and realised quarticity RQ; target: next-day RV in levels, full sample
  - SHAR Patton & Sheppard (2015): $\mathrm{RV}_{t+1} = \beta_0 + \beta^+ \mathrm{RV}_t^+ + \beta^- \mathrm{RV}_t^- + \beta_w \mathrm{RV}_t^{(w)} + \beta_m \mathrm{RV}_t^{(m)} + \varepsilon_{t+1}$, with β^+, β^- the weights of the positive and negative semivariance
  - HARQ Bollerslev, Patton & Quaedvlieg (2016): HAR plus the term $\beta_Q \, \mathrm{RV}_t (\mathrm{RQ}_t^{1/2} - \overline{\mathrm{RQ}^{1/2}})$ (Recall, the HAR Family (B1, B2))
  - V: Newey–West (5 lags) covariance matrix of the estimates; V_{++}, V_{--}: variances of $\hat{\beta}^+$, $\hat{\beta}^-$; V_{+-}: their covariance
- **Tasks**
  (a) Estimate HAR Corsi (2009) by OLS and report the three slopes with HAC standard errors.
  (b) Estimate SHAR and test $H_0 : \beta^- = \beta^+$ against $\beta^- > \beta^+$ with $t = (\hat{\beta}^- - \hat{\beta}^+)/\mathrm{se}(\hat{\beta}^- - \hat{\beta}^+)$, the HAC standard error of a difference of Newey–West (HAC) Standard Errors (B1, B2, B3).
  (c) Estimate HARQ and test $H_0 : \beta_Q = 0$ against $\beta_Q < 0$; plot β^+, β^- and β_Q with 95% HAC intervals.
- **Report**: estimates, t, p, decisions, the coefficient chart
- **Interpretation**: do the two effects of the literature hold for this stock?
- Notebook: section B2

Back Exercise list

B2 [Solved]: Solution (1/2) — Estimates and Tests

- (a) HAR (2 764 days): $\beta_d = 0.367$ (0.080), $\beta_w = 0.494$ (0.149), $\beta_m = -0.026$ (0.095)
- (b) SHAR: $\beta^+ = 0.401$ (0.252), $\beta^- = 0.324$ (0.236); difference -0.078 (from unrounded estimates), $t = -0.17$, one-sided $p = 0.566$: do not reject H_0
- (c) HARQ: $\hat{\beta}_Q = -35.062$ (15.904), $t = -2.20 < -1.645$: reject H_0 at 5% one-sided

B2 [Solved]: Solution (2/2) — Chart and Interpretation



- Chart: the two SHAR intervals are wide and overlap (the semivariances are highly correlated for one stock); the HARQ interval lies below zero
- Interpretation: on noisy days HARQ trusts the day's RV less, and this holds for JPM; the good/bad asymmetry is a panel result Patton & Sheppard (2015), not detectable for JPM alone

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B3 [Solved]: Sig-LK vs log-HAR for JPM, One Day Ahead — Task

- **Question**: does the signature method of the lecture (Sig-LK) forecast JPM's next-day RV better than log-HAR?
- **Inputs**
  - VOLARE Cipollini et al. (2026), JPM (JPMorgan Chase), 2 Jan 2015 – 30 Jan 2026, trading days only; five-minute RV, daily open and close prices
  - Design of Chapter 20 (Sig-LK, Features and Kernel Weights, Sig-LK, Effective Sample Size and the Two-Step LASSO): rolling window of 1000 days, re-estimation every 5 days, path (time, $\ln \mathrm{RV}$, cumulative r) over 22 days, depth 3, two-step LASSO with BIC, inverse-temperature scale $c = 0.5$
- **Tasks**
  (a) Produce one-day forecasts of Sig-LK and log-HAR from 27 Jan 2020 to 29 Jan 2026 (1 511 days) and compute the mean QLIKE of each.
  (b) Compute the ratio from unrounded mean losses and its 95% CI with the stationary bootstrap (mean block length 10, 1000 replications).
  (c) Run the DM test of Recall, Comparing Two Forecasts (B3) (HAC variance, bandwidth $\max(h, \lceil T^{1/3} \rceil)$) and the conditional GW test with instruments $(1, d_{t-1})$ Giacomini & White (2006).
  (d) Plot the cumulative loss difference $\sum_{s \le t} d_s$.
- **Report**: losses, ratio and CI, t, p, decisions, the chart
- **Interpretation**: is there any evidence that the signature adds value for JPM?
- Notebook: section B3

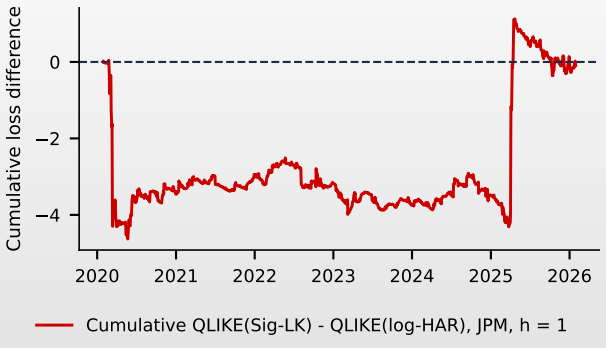
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B3 [Solved]: Solution (1/2) — Losses and Tests

- (a) Mean QLIKE: Sig-LK 0.1726, log-HAR 0.1727; on average 4.2 regressors selected, effective sample size 851 of 1000
- (b) Ratio 1.000; bootstrap 95% CI $[0.956, 1.047]$ contains 1
- (c) DM: $d_t = \mathrm{QLIKE}_t^{\text{Sig-LK}} - \mathrm{QLIKE}_t^{\text{log-HAR}}$, $t = \bar{d}/\sqrt{\hat{\Omega}_{\mathrm{NW}}/T} = -0.02$, one-sided $p = 0.493$: do not reject equal accuracy
- GW: Wald statistic $2.85 \sim \chi^2(2)$, $p = 0.241$: the instruments $(1, d_{t-1})$ give no significant evidence that the loss difference is predictable

B3 [Solved]: Solution (2/2) — Cumulative Loss Difference

- (d) Upward segments: log-HAR better; downward segments: Sig-LK better; the largest moves sit in March 2020 and April 2025
- A handful of stress days decides the sign of the total: report the cumulative path, not only the mean
- Interpretation: no evidence; the two forecasts are practically identical, and the CI rules out gains or losses larger than about 5%



B4 [Solved]: Holm and BH, Step by Step — Task

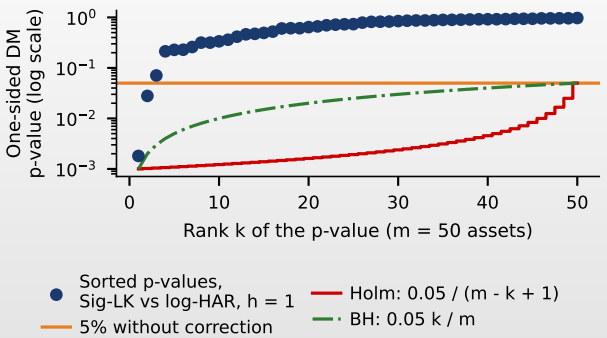
- **Question**: after correcting for 50 tests, on how many assets does Sig-LK beat log-HAR one day ahead?
- **Inputs**
  - Per-asset DM p-values from Chapter 20 (provided with the notebook): one-sided DM p-values for "Sig-LK better than log-HAR" at $h = 1$ (QLIKE), one per asset; family = the 50 assets
  - The five FX pairs, sorted: EURUSD 0.071, USDJPY 0.213, USDCAD 0.336, AUDUSD 0.718, GBPUSD 0.817
  - m: number of tests; Holm (FWER 5%) and BH (FDR 5%) Benjamini & Hochberg (1995) as in Recall, Many Tests at Once (B4)
- **Tasks**
  (a) Apply Holm to the five FX p-values ($m = 5$).
  (b) Apply BH to the same five.
  (c) Repeat both procedures for all 50 assets, count the rejections and plot the sorted p-values against the three thresholds.
- **Report**: rejections raw, Holm, BH; the chart
- **Interpretation**
  - what may we claim?
  - about how many assets?
- Notebook: section B4

B4 [Solved]: Solution (1/2) — Steps

- (a) Holm, $m = 5$: $p_{(\mathbf{1})} = 0.071$ vs $0.05/5 = 0.01$: no rejection, stop; Holm-adjusted p: 0.354, 0.852, then 1 (from unrounded p-values)
- (b) BH: thresholds $0.01, 0.02, 0.03, 0.04, 0.05$; every $p_{(k)}$ is above its threshold: no rejection; the smallest BH-adjusted p is 0.354 (from unrounded p-values)
- (c) 50 assets: 2 raw rejections; Holm: the smallest, ES with $p = 0.002$, is above $0.05/50 = 0.001$: stop, 0 rejections; BH: $p_{(\mathbf{1})} > 0.001$ and $p_{(\mathbf{2})} = 0.028 > 0.002$, no k qualifies: 0

B4 [Solved]: Solution (2/2) — Chart and Interpretation

- Chart: only the lowest points cross the 5% line; none reaches the Holm staircase or the BH line
- Holm is valid under any dependence; BH needs independent or positively dependent *p*-values (PRDS: Positive Regression Dependence on a Subset) Benjamini & Yekutieli (2001)
- Interpretation: no asset-level improvement survives either correction; ES is at most a lead for a new, pre-registered study



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C1: Data and Helper Convention

- Use the JPM RV series from the setup. The last 30% of observations form the evaluation split for both the candidate and log-HAR; the earlier part is the training set.
- `sig_path(X, depth)` takes the rows of an array as successive points of a piecewise-linear path. It returns the flattened signature levels 1 through depth, without level 0 and without adding any channel.
- Assess both the program and its claimed conclusion: a comparison mentioned in prose must be supported by forecasts and losses computed on the same evaluation rows.

C1 [Proposed]: Critique an AI Answer — the Answer

- **Prompt** sent to an AI assistant: "rv holds the daily 5-minute RV of JPM from the setup data. Build the signature LASSO of Gu et al. for next-day log RV and evaluate it with QLIKE on the last 30% of the sample."
- **Answer**: the code below and the conclusion

```
lrv = np.log(rv.values)
S = np.array([sig_path(lrv[t-21:t+1, None], depth=3)   # 22-day window
              for t in range(21, len(lrv) - 1)])
y = lrv[22:]                                            # next-day log RV
lam = LassoCV(cv=5).fit(S, y).alpha_                    # choose lambda
cut = int(0.7 * len(y))
m = Lasso(alpha=lam).fit(S[:cut], y[:cut])
f = np.exp(m.predict(S[cut:]))
qlike = lambda a, b: a / b - np.log(a / b) - 1
print(qlike(f, np.exp(y[cut:])).mean())
```

- **Conclusion of the AI**: "The signature LASSO has a lower QLIKE than HAR on the test set, so path signatures capture the shape of volatility and improve the forecast."

Back Exercise list

C1 [Proposed]: Critique an AI Answer — Task

Solution discussed in class

- **Question**: which errors in the answer would change the conclusion?
- **Model**: A1 (time augmentation), A2 (QLIKE), B3 (evaluation without look-ahead); the code contains **at least three** errors
- **Tasks**
  (a) Find at least three errors; for each, name the line and explain in one sentence why it is wrong.
  (b) For each error, describe a check that reveals it without knowing the answer (a unit test, a timing test, a known case).
  (c) Assess the exponential back-transform and whether the code is the two-step weighted method of Gu et al.
  (d) Write the corrected code and compare it with log-HAR on the same split, with a DM test.
  (e) Log each error as in `AI_ERRORS.md`: request, answer, error, how found, correction.
- **Interpretation**: does the conclusion follow even if the code were correct?
- AI code usually runs without an error message: the mistakes are in the method, not in the syntax

C1: Design Choices and Evidence

- Use a fixed training fit or a rolling fit, equal or kernel weights, and a penalty selector fitted only with information available at each forecast. State the channels and justify every design choice in one sentence.
- Report an error table (line, reason, diagnostic check), the corrected code and the entries in AI_ERRORS.md.
- On the same evaluation rows, report both mean QLIKE losses, the DM statistic, its one-sided p-value and the direction of the alternative. Explain whether these results support the original claim.

Extension Packs: What Each One Needs

Pack	Exercises	Prerequisites
1. Signature algebra	A4, A5 [Solved], A6 (required)	A1
2. HAR as an AR(22)	A7 [Solved], A8	B2, Wald tests
3. Losses and weights	A9, A10	A2, A3
4. Other assets	B5 (ES), B6 (CL), B7 (ES, $h = 5$)	B1, B2, B3
5. Multiple testing at one month	B8	B4
6. Project seed	C2, a level-aware kernel	B3, Chapter 20 (kernel weights)

- Each exercise has its own notebook section with starter code (an empty code cell for proposed ones); within a pack, work in the order given
- Proposed exercises name their model: the solved exercise that uses the same technique

A4 [Proposed]: Signature and Shuffle Identity — Task

Solution discussed in class

- **Questions**
  - what is the signature of a three-segment path in the plane?
  - what does signed area suggest about the order of moves?
- **Model**: A1
- **Inputs**: A path in $\mathbb{R}^2$ through $(0,0) \to (2,1) \to (3,-1) \to (4,2)$, linear between points
- **Tasks**
  (a) Write the increments and compute level 1.
  (b) Compute level 2 with the formula of A1 (squares plus ordered pairs).
  (c) Verify $S^1 S^2 = S^{12} + S^{21}$ and compute the Lévy area A^{12}.
- **Report**: $S^{(1)}$, $S^{(2)}$, A^{12}; **Interpretation**: what does the sign of the area say about the tendency of channel 1 to move before channel 2? Why does it not imply strict precedence?
- Notebook: section A4

A5 [Solved]: Chen's Identity and Path Reversal — Task

- **Questions**
  - what is the signature of a path run there and back?
  - why does this matter for time augmentation?
- **Inputs**
  - Path $X : (0,0) \to (1,0) \to (1,2)$ from the lecture, with $S^{(1)} = (1,2)$ and $S^{(2)} = \left[\begin{smallmatrix} 1/2 & 2 \\ 0 & 2 \end{smallmatrix}\right]$
  - $\overleftarrow{X}$ is the same path run backwards: $(1,2) \to (1,0) \to (0,0)$
- **Tasks**
  (a) Compute levels 1 and 2 of $\mathrm{Sig}(\overleftarrow{X})$ with the A1 formula.
  (b) Apply Chen's identity at level 2 (Chen's Identity and the Shuffle Identity) with $A = \mathrm{Sig}(X)$ and $B = \mathrm{Sig}(\overleftarrow{X})$, to compute level 2 of $X * \overleftarrow{X}$ (X followed by $\overleftarrow{X}$).
  (c) State what the result implies for the uniqueness of the signature.
- **Report**: the two signatures and level 2 of the concatenation; **Interpretation**: why do we add a time channel in practice?
- Notebook: section A5

Back Exercise list

A5 [Solved]: Solution

- (a) Reversed increments $(0,-2)$, $(-1,0)$: level 1 is $(-1,-2) = -S^{(1)}$; level 2: squares $\left[\begin{smallmatrix} 1/2 & 0 \\ 0 & 2 \end{smallmatrix}\right]$ plus the pair $\left[\begin{smallmatrix} 0 & 0 \\ 2 & 0 \end{smallmatrix}\right]$, so $\overleftarrow{S}^{(2)} = \left[\begin{smallmatrix} 1/2 & 0 \\ 2 & 2 \end{smallmatrix}\right]$
- (b) Chen: $\left[\begin{smallmatrix} 1/2 & 2 \\ 0 & 2 \end{smallmatrix}\right] + \left[\begin{smallmatrix} -1 & -2 \\ -2 & -4 \end{smallmatrix}\right] + \left[\begin{smallmatrix} 1/2 & 0 \\ 2 & 2 \end{smallmatrix}\right] = \left[\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}\right]$; level 1 is $S^{(1)} - S^{(1)} = 0$ as well
- (c) There and back is "tree-like": its signature is that of the constant path; the signature identifies a path only up to such pieces Hambly & Lyons (2010)
- Interpretation: a strictly increasing time channel cannot go back, so tree-like pieces are impossible and the signature determines the path up to its starting point: we time-augment and supply the level separately

A6 [Proposed]: Lévy Area of a Loop — Task

Solution discussed in class

- **Question**: what does level 2 of the signature see in a closed loop that level 1 misses?
- **Model**: A5
- **Inputs**: The unit square $(0,0) \to (1,0) \to (1,1) \to (0,1) \to (0,0)$, four linear segments
- **Tasks**
  (a) Compute level 1 and the Lévy area A^{12} for the square run anticlockwise.
  (b) Repeat for the square run clockwise, and for the square translated by $(5,5)$.
  (c) Explain why level 1 cannot distinguish the loop from a constant path while level 2 can.
- **Report**: three values of $S^{(1)}$ and A^{12}; **Interpretation**: which invariance does the translation illustrate?
- Notebook: section A6

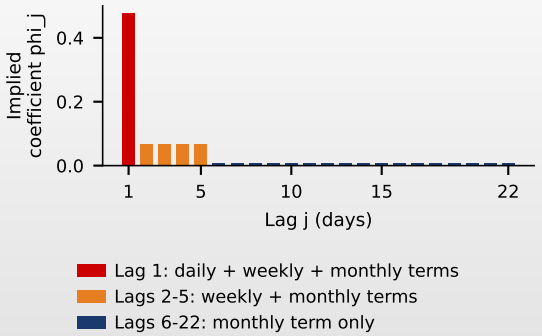
A7 [Solved]: HAR as a Restricted AR(22) — Task

- **Question**: which 22 lag coefficients does a HAR model for $\ln \mathrm{RV}$ imply for JPM?
- **Inputs**
  - VOLARE Cipollini et al. (2026), JPM (JPMorgan Chase), 2 Jan 2015 – 30 Jan 2026, trading days only
  - HAR with *averages of logs*: $\ell_t = \ln \mathrm{RV}_t$, $\ell_t^{(w)} = \frac{1}{5}\sum_{j=0}^{4} \ell_{t-j}$, $\ell_t^{(m)} = \frac{1}{22}\sum_{j=0}^{21} \ell_{t-j}$
  - The lecture's log-HAR uses logs of averages, $\ln \mathrm{RV}_t^{(w)} \neq \ell_t^{(w)}$ (Jensen), and has no exact AR form
  - Full sample (2 764 days), OLS, Newey–West (5 lags) s.e.: $\hat{\beta}_d = 0.408$ (0.027), $\hat{\beta}_w = 0.299$ (0.046), $\hat{\beta}_m = 0.196$ (0.040)
- **Tasks**
  (a) Substitute $\ell_t^{(w)}$ and $\ell_t^{(m)}$ into the HAR equation and collect the coefficient ϕ_j of each lag ℓ_{t-j+1}, $j = 1, \ldots, 22$.
  (b) Compute ϕ_1, $\phi_2 = \cdots = \phi_5$ and $\phi_6 = \cdots = \phi_{22}$ from the JPM estimates and plot them.
  (c) Compute the sum $\sum_{j=1}^{22} \phi_j$.
- **Report**: the three distinct values, the sum, the chart
- **Interpretation**
  - what does the sum measure?
  - is the process stationary?
- Notebook: section A7

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A7 [Solved]: Solution

- (a) Using unrounded fitted coefficients: $\beta_w \ell_t^{(w)} = \sum_{j=1}^{5} (\beta_w/5)\ell_{t-j+1}$ and $\beta_m \ell_t^{(m)} = \sum_{j=1}^{22} (\beta_m/22)\ell_{t-j+1}$: lag 1 collects all three terms, lags 2–5 the weekly and monthly ones, lags 6–22 only the monthly one
- (b) $\phi_1 = 0.408 + 0.299/5 + 0.196/22 = 0.476$; $\phi_{2..5} = 0.0688$; $\phi_{6..22} = 0.0089$
- (c) $\sum_j \phi_j = \beta_d + \beta_w + \beta_m = 0.903$
- Interpretation: the sum is the persistence of log RV; all $\phi_j \geq 0$ and the sum is below 1, so the process is stationary, with a slow, long-memory-like decay



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A8 [Proposed]: Testing the HAR Restrictions — Task

Solution discussed in class

- **Question**: how do we test the restrictions that HAR imposes on an unrestricted AR(22)?
- **Model**: A7
- **Inputs**
  - The unrestricted AR(22) for $\ell_t = \ln \mathrm{RV}_t$ has an intercept and slopes $\phi_1, \ldots, \phi_{22}$; HAR with averages of logs (A7) is its restricted version
  - ϕ is the linear projection of ℓ_{t+1} on its 22 lags; weakly dependent data with finite fourth moments; errors may be heteroskedastic and serially correlated
- **Tasks**
  (a) Count the linear restrictions and write them as $R\phi = 0$.
  (b) Write the Wald statistic with an HAC covariance and its distribution under H_0.
  (c) State the 5% critical value.
- **Interpretation**: if the test rejects in a long sample, should we drop HAR for forecasting?
- Report the restriction matrix, Wald statistic, degrees of freedom and critical value.
- Notebook: section A8

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A9 [Proposed]: QLIKE With Its Arguments Swapped — Task

Solution discussed in class

- **Question**: which forecasts does a QLIKE with swapped arguments reward?
- **Model**: A2
- **Inputs**: Assume $\mathrm{RV} > 0$ and $\mathbb{E}_t[1/\mathrm{RV}] < \infty$. A program computes $L(\mathrm{RV}, F) = F/\mathrm{RV} - \ln(F/\mathrm{RV}) - 1$, QLIKE with the arguments swapped
- **Tasks**
  (a) Find the forecast $F^\dagger$ that minimises $\mathbb{E}_t[L(\mathrm{RV}, F)]$.
  (b) Compare $F^\dagger$ with $\mathbb{E}_t[\mathrm{RV}]$ using Jensen's inequality.
  (c) Say which models the swapped loss favours.
- **Interpretation**: why can this bug pass elementary checks?
- Report the first-order condition, minimiser and comparison with the arithmetic-mean target.
- Notebook: section A9

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A10 [Proposed]: Limits of the Temperature — Task

Solution discussed in class

- **Questions**
  - what happens to the kernel weights at the two extreme temperatures?
  - why scale by the median distance?
- **Model**: A3
- **Inputs**: n past windows, distances $\delta_\tau > 0$, temperature $\gamma = c/\mathrm{median}(\delta)$ as in Sig-LK, Features and Kernel Weights
- **Tasks**
  (a) Find the limits of w and n_{eff} when $c \to 0$.
  (b) Find the limits when $c \to \infty$ (assume a unique nearest window).
  (c) Show that multiplying all distances by $k > 0$ leaves the weights unchanged.
- **Interpretation**: why is the median scaling needed to compare c across 50 assets?
- Report both weight-vector limits and effective sample sizes, with their interpretation.
- Notebook: section A10

B5 [Proposed]: The Lévy Area for E-mini S&P 500 Futures — Task

Solution discussed in class

- **Question**: does the Lévy area predict next month's volatility of E-mini S&P 500 futures (ES)?
- **Model**: B1
- **Inputs**: VOLARE futures data, 28 Sep 2009 – 30 Jan 2026, ES; five-minute RV, open and close prices; target, regressors and 22 Newey–West lags as in B1
- **Tasks**
  (a) Repeat steps (a)–(d) of B1 for ES.
  (b) Test the coefficient of A_t at 1% and 5%, two-sided.
  (c) Compare the R^2 gain with the one for JPM.
- **Report**: the regression table, both significance decisions, the change in R-squared and the binned scatter.
- **Interpretation**
  - what does a positive coefficient say about windows with a negative area?
  - can the sign alone label them leverage episodes?
- Notebook: section B5

B6 [Proposed]: HAR, SHAR and HARQ for Crude Oil — Task

Solution discussed in class

- **Question**: do the SHAR asymmetry and the HARQ effect hold for crude oil futures (CL)?
- **Model**: B2
- **Inputs**: VOLARE futures data, 28 Sep 2009 – 30 Jan 2026, CL; five-minute RV, semivariances and realised quarticity; models, tests and Newey–West (5 lags) as in B2
- **Tasks**
  (a) Estimate SHAR and test $\beta^- > \beta^+$ as in B2(b).
  (b) Estimate HARQ and test $\beta_Q < 0$ as in B2(c); plot the coefficients with 95% HAC intervals.
  (c) Compare both results with JPM.
- **Report**: estimates with HAC s.e., two one-sided tests and 5% decisions, the coefficient chart, and comparison with JPM.
- **Interpretation**: is the asymmetry stronger or weaker than for JPM, and what economic mechanism could explain either outcome?
- Notebook: section B6

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B7 [Proposed]: Sig-LK vs log-HAR for ES, One Week Ahead — Task

Solution discussed in class

- **Question**: does Sig-LK beat log-HAR for ES when the target is the average RV of the next five days?
- **Model**: B3
- **Inputs**: VOLARE futures data, 28 Sep 2009 – 30 Jan 2026, ES; design of B3 with $h = 5$ and $c = 1.0$ (futures validation block)
- **Tasks**
  (a) Produce forecasts at every daily origin of the five-day average RV, using B3 and the futures validation cutoff; evaluation origins 7 Oct 2014 to 23 Jan 2026.
  (b) Compute the QLIKE ratio and its stationary-bootstrap 95% CI (mean block length 10).
  (c) Run the DM test (bandwidth at least $h = 5$) and the conditional GW test using instruments $(1, d_{t-\mathbf{5}})$; plot the cumulative loss difference.
- **Report**: both mean losses, their ratio and interval, the DM/GW statistics and p-values, and the cumulative-loss plot.
- **Interpretation**
  - is the CI asymmetric, and which features of the losses could explain its shape?
  - what does it imply for the power of DM?
- Notebook: section B7

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B8 [Proposed]: Multiple Testing at One Month Against HAR — Task

Solution discussed in class

- **Question**: on how many assets does Sig-LK beat HAR one month ahead after correcting for 50 tests?
- **Model**: B4
- **Inputs**: Chapter 20 results: 50 one-sided DM p-values for "Sig-LK better than HAR" at $h = 22$ (QLIKE)
- **Tasks**
  (a) Count the rejections at 5% without correction.
  (b) Apply Holm and BH as in B4, count the rejections and name the asset with the smallest p-value.
  (c) Compare with the mean across assets of the per-asset QLIKE ratios of Sig-LK relative to HAR at $h = 22$ (1.334).
- **Report**: sorted p-values, both thresholds and decisions, three rejection counts, and the named asset.
- **Interpretation**: how can Sig-LK beat HAR on several assets and still lose on average?
- Notebook: section B8

C2 [Proposed]: A Level-Aware Signature Kernel — Task

Solution discussed in class

- **Question**: does a kernel that also sees the *level* of volatility make the weights useful in a crash?
- **Inputs**
  - The kernel of the lecture is translation-invariant, so it cannot recognise a high-volatility regime (Chapter 20, weights on 16 March 2020); data: VOLARE, the 40 stocks, 2015–2026
  - Idea: prepend a **basepoint** Morrill et al. (2020): the path starts at the training mean of $\ln \mathrm{RV}$ and jumps to the first value of the window; by Chen, $\mathrm{Sig} = \exp_{\otimes}(v) \otimes \mathrm{Sig}(X)$ with v the standardised level jump and $\exp_{\otimes}(v) = (1, v, \frac{1}{2}v^{\otimes 2}, \dots)$ the signature of that single jump segment
- **Tasks**
  (a) Write a one-page pre-registration: 40 stocks, $h = 1$, the lecture design, $c = 0.5$, QLIKE, DM with Holm, the COVID-19 subperiod reported separately.
  (b) Implement the basepoint kernel (use the Chen update of the notebook) with the same features as Sig-LK, and forecast.
  (c) Compare with Sig-LK and log-HAR: mean QLIKE ratios (all days and COVID-19), wins over log-HAR raw and after Holm.
- **Interpretation**: if seeing the level does not help, where would you look next (features, target, horizon)?
- Notebook: section C2

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C2: The Basepoint Jump

- Use channel order (time, log RV, cumulative return) and prepend $v = (0, (\ln \mathrm{RV}_{\mathrm{first}} - \mu_{\mathrm{train}})/s_{\mathrm{train}}, 0)$. Here the mean and standard deviation use log RV at the 1000 training origins, without future observations.
- Keep the ordinary path scaling from B3. This additional jump uses the standard deviation of levels, not of increments. Its time and return components are zero.
- Unlike the zero basepoint illustration in the course, this basepoint is centred on the training mean. It measures whether the starting volatility level is unusually high relative to that training sample.
- Report the pre-registration, QLIKE ratio tables for all days and COVID-19, raw and Holm-adjusted win counts, and a comparison of weights on 16 March 2020.

Further Study on Quantinar

Advanced courses on quantinar.com

- Statistics of Financial Markets
- Applied Time Series Analysis with Python

Code and data

- Seminar notebook (Colab); Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_20; data: volare.unime.it

Acronyms used in this material (1/2)

- **AI**: Artificial Intelligence
- **AR**: AutoRegressive (model)
- **AUD**: Australian Dollar (ISO 4217 code)
- **AUDUSD**: Australian dollar / US dollar exchange rate (FX pair code)
- **BH**: Benjamini–Hochberg (procedure)
- **BIC**: Bayesian Information Criterion
- **CAD**: Canadian Dollar (ISO 4217 code)
- **CI**: Confidence Interval
- **CL**: Crude oil (WTI light sweet) futures, NYMEX ticker
- **COVID-19**: COronaVIrus Disease 2019
- **DM**: Diebold–Mariano (test)
- **ES**: E-mini S&P 500 futures (CME ticker)
- **EURUSD**: Euro / US dollar exchange rate (FX pair code)
- **FDR**: False Discovery Rate
- **FWER**: Family-Wise Error Rate
- **FX**: Foreign eXchange
- **GBPUSD**: British pound / US dollar exchange rate (FX pair code)
- **GW**: Giacomini–White (test)
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)

Acronyms used in this material (2/2)

- **HAR**: Heterogeneous AutoRegressive (model)
- **HARQ**: HAR model with Realised Quarticity
- **JPM**: JPMorgan Chase & Co. (NYSE ticker)
- **LASSO**: Least Absolute Shrinkage and Selection Operator
- **MA**: Moving Average
- **NW**: Newey–West (standard errors)
- **OLS**: Ordinary Least Squares
- **PRDS**: Positive Regression Dependence on a Subset (of p-values)
- **QLIKE**: Quasi-LIKElihood (loss function)
- **RQ**: Realised Quarticity
- **RV**: Realised Variance
- **SHAR**: Semivariance HAR model
- **US**: United States
- **USDCAD**: US dollar / Canadian dollar exchange rate (FX pair code)
- **USDJPY**: US dollar / Japanese yen exchange rate (FX pair code)
- **VOLARE**: VOLatility Archive for Realized Estimates (database)

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